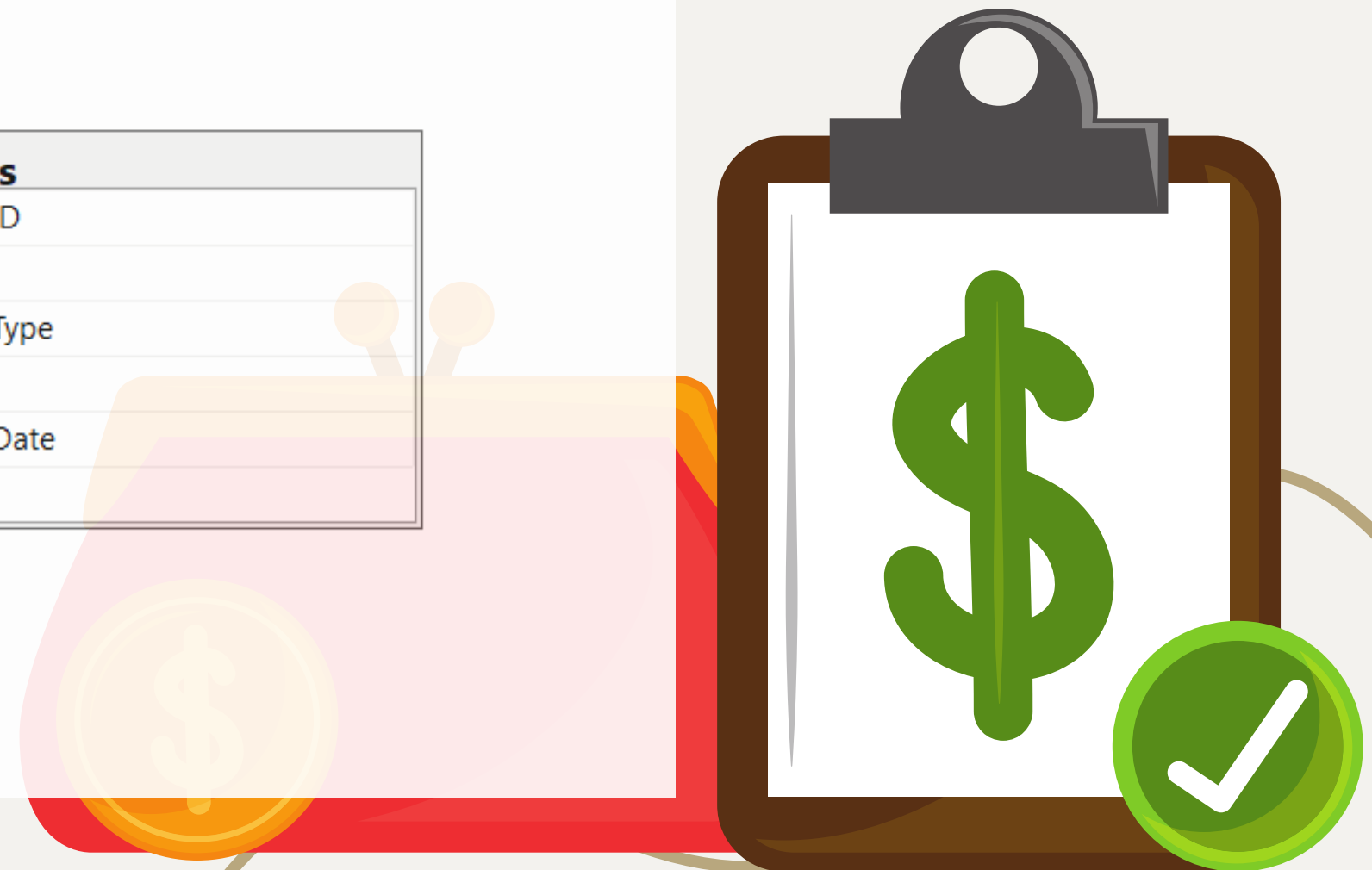
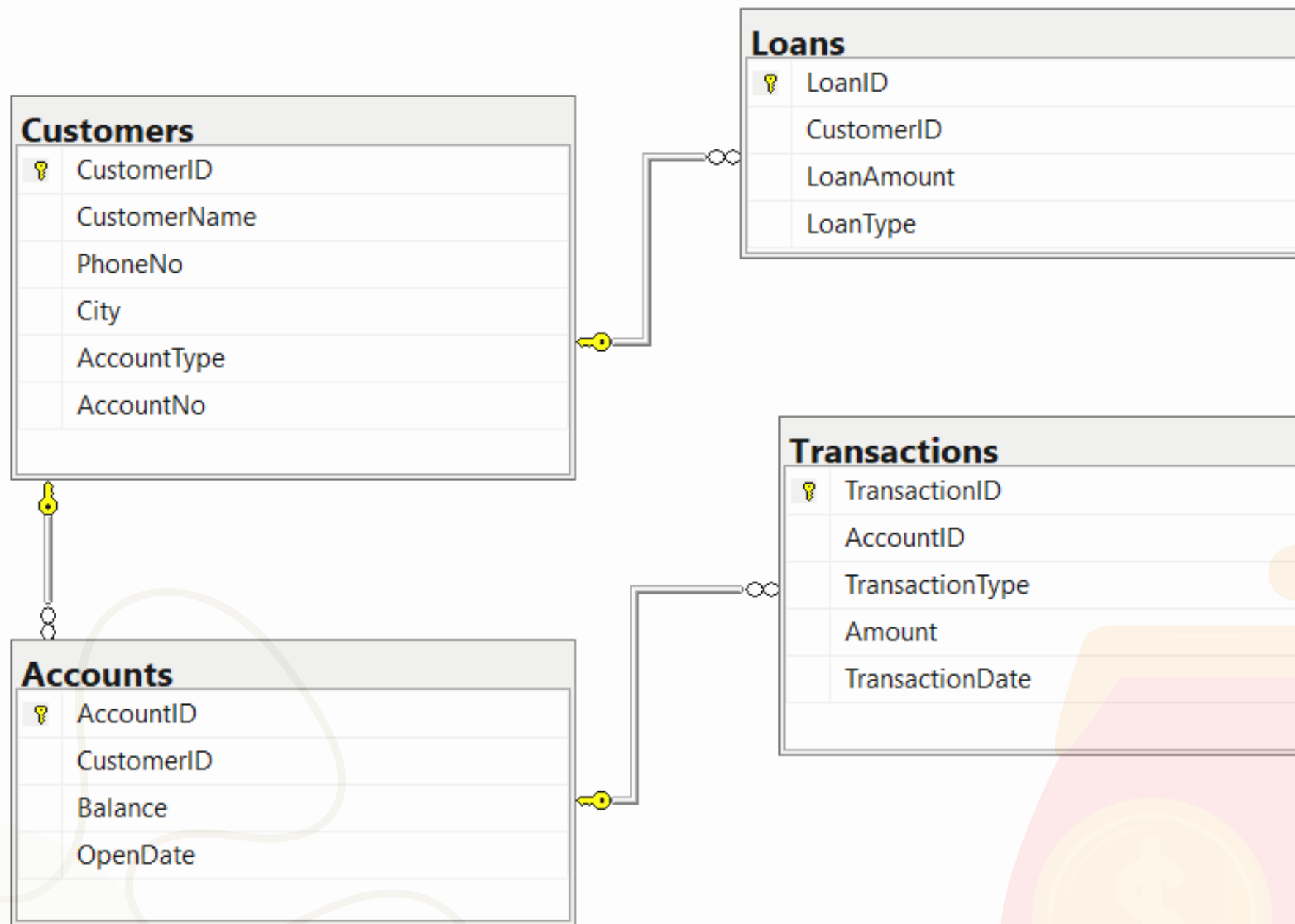




HELLO

My name is Ashraf ul Ahmad , in this project
I have used SQL queries to solve questions
related to
Banking Management System

SCHEMA DIAGRAM



1.Display customer names, account numbers, and account balances using INNER JOIN.

```
select c.CustomerName , a.AccountID , a.Balance
from Accounts a
inner join Customers c on a.CustomerID = c.CustomerID;
```



	CustomerName	AccountID	Balance
1	Rahul Sharma	1001	55000
2	Sneha Patil	1002	120000
3	Aman Verma	1003	35000
4	Priya Singh	1004	98000
5	Karan Mehta	1005	75000
6	Neha Joshi	1006	150000
7	Rohit Kumar	1007	42000
8	Pooja Sharma	1008	88000
9	Vivek Shah	1009	200000
10	Anjali Verma	1010	67000

2. Find the top 3 customers with the highest account balances.

```
select top (3) c.CustomerName , a.Balance  
from Accounts a  
inner join Customers c on a.CustomerID = c.CustomerID  
order by Balance desc;
```



A stylized illustration of a hand holding a green banknote, positioned on the left side of the slide. The hand is light orange, and the banknote is green with yellow circles. A red square is partially visible on the far left.

Results		Messages
	CustomerName	Balance
1	Vivek Shah	200000
2	Neha Joshi	150000
3	Sneha Patil	120000

3. Show all customers who have taken loans along with loan amount and loan type.

```
select c.CustomerName , l.LoanAmount , l.LoanType
from Loans l
left join Customers c on l.CustomerID = c.CustomerID;
```



	CustomerName	LoanAmount	LoanType
1	Rahul Sharma	500000	Home Loan
2	Sneha Patil	200000	Car Loan
3	Priya Singh	100000	Education Loan
4	Karan Mehta	300000	Business Loan
5	Neha Joshi	150000	Personal Loan
6	Pooja Sharma	250000	Home Loan
7	Vivek Shah	400000	Business Loan
8	Anjali Verma	180000	Car Loan

4. Find the total deposited amount and total withdrawn amount separately.

```
select TransactionType, sum(Amount) as Total  
from Transactions  
group by TransactionType;
```



Results Messages		
	TransactionType	Total
1	Deposit	121000
2	Withdraw	35000

5. Display customer-wise total transaction amount using GROUP BY.

```
select c.CustomerName , sum(t.Amount) as Total_Amount
from Transactions t
join Accounts a on t.AccountID = a.AccountID
join Customers c on a.CustomerID = c.CustomerID
group by c.CustomerName;
```

Results Messages

	CustomerName	Total_Amount
1	Aman Verma	8000
2	Anjali Verma	2000
3	Karan Mehta	15500
4	Neha Joshi	7000
5	Pooja Sharma	4500
6	Priya Singh	15000
7	Rahul Sharma	15000
8	Rohit Kumar	24000
9	Sneha Patil	35000
10	Vivek Shah	30000

6. Find customers whose balances are greater than the average bank balance.

```
select c.CustomerName , a.Balance
from Accounts a
join Customers c on a.CustomerID = c.CustomerID
where a.Balance > (select avg(Balance) from Accounts);
```



Results Messages		
	CustomerName	Balance
1	Sneha Patil	120000
2	Priya Singh	98000
3	Neha Joshi	150000
4	Vivek Shah	200000

7. Show the highest transaction amount performed by each customer.

```
select c.CustomerName , Max(t.Amount) as Highest_Amount
from Transactions t
join Accounts a on t.AccountID = a.AccountID
join Customers c on a.CustomerID = c.CustomerID
group by c.CustomerName;
```

Results Messages		
	CustomerName	Highest_Amount
1	Aman Verma	5000
2	Anjali Verma	2000
3	Karan Mehta	12000
4	Neha Joshi	7000
5	Pooja Sharma	4500
6	Priya Singh	15000
7	Rahul Sharma	10000
8	Rohit Kumar	15000
9	Sneha Patil	25000
10	Vivek Shah	30000

8. Display all customers who have not taken any loans using LEFT JOIN.

```
select c.CustomerName
from Customers c
left join Loans l on l.CustomerID = c.CustomerID
where l.LoanAmount is null;
```



Results		Messages			
	CustomerName				
1	Aman Verma				
2	Rohit Kumar				

9. Find the total number of transactions performed by each customer.

```
select c.CustomerName , count(t.AccountID) as No_of_Transaction
from Transactions t
join Accounts a on t.AccountID = a.AccountID
join Customers c on a.CustomerID = c.CustomerID
group by c.CustomerName;
```

Results Messages		
	CustomerName	No_of_Transaction
1	Aman Verma	2
2	Anjali Verma	1
3	Karan Mehta	2
4	Neha Joshi	1
5	Pooja Sharma	1
6	Priya Singh	1
7	Rahul Sharma	2
8	Rohit Kumar	2
9	Sneha Patil	2
10	Vivek Shah	1

10. Rank customers based on their account balances using RANK() window function.

```
select c.CustomerName , a.Balance ,  
rank() over(order by a.Balance desc) as Rank_Number  
from Accounts a  
join Customers c on a.CustomerID = c.CustomerID;
```



	CustomerName	Balance	Rank_Number
1	Vivek Shah	200000	1
2	Neha Joshi	150000	2
3	Sneha Patil	120000	3
4	Priya Singh	98000	4
5	Pooja Sharma	88000	5
6	Karan Mehta	75000	6
7	Anjali Verma	67000	7
8	Rahul Sharma	55000	8
9	Rohit Kumar	42000	9
10	Aman Verma	35000	10

11. Display dense ranking of customers according to balance using DENSE_RANK()

```
select c.CustomerName , a.Balance ,  
dense_rank() over(order by a.Balance desc) as Dense_Rank_No  
from Accounts a  
join Customers c on a.CustomerID = c.CustomerID;
```

Results Messages			
	CustomerName	Balance	Dense_Rank_No
1	Vivek Shah	200000	1
2	Neha Joshi	150000	2
3	Sneha Patil	120000	3
4	Priya Singh	98000	4
5	Pooja Sharma	88000	5
6	Karan Mehta	75000	6
7	Anjali Verma	67000	7
8	Rahul Sharma	55000	8
9	Rohit Kumar	42000	9
10	Aman Verma	35000	10



12. Show previous transaction amount using LAG() function.

```
select TransactionID , AccountID , Amount , TransactionType ,  
lag(Amount,1,0) over(partition by AccountID order by TransactionDate) as Previous_Amount  
from Transactions;
```

	TransactionID	AccountID	Amount	TransactionType	Previous_Amount
5	4	1003	3000	Withdraw	0
6	13	1003	5000	Deposit	3000
7	5	1004	15000	Deposit	0
8	6	1005	12000	Deposit	0
9	14	1005	3500	Withdraw	12000
10	7	1006	7000	Withdraw	0
11	8	1007	9000	Deposit	0
12	15	1007	15000	Deposit	9000
13	9	1008	4500	Withdraw	0
14	10	1009	30000	Deposit	0
15	11	1010	2000	Withdraw	0



13. Show next transaction amount using LEAD() function

```
select TransactionID , AccountID , Amount ,TransactionType,  
lead(Amount,1,0) over(partition by AccountID order by TransactionDate) as Next_Amount  
from Transactions;
```

	TransactionID	AccountID	Amount	TransactionType	Next_Amount
4	12	1002	10000	Withdraw	0
5	4	1003	3000	Withdraw	5000
6	13	1003	5000	Deposit	0
7	5	1004	15000	Deposit	0
8	6	1005	12000	Deposit	3500
9	14	1005	3500	Withdraw	0
10	7	1006	7000	Withdraw	0
11	8	1007	9000	Deposit	15000
12	15	1007	15000	Deposit	0
13	9	1008	4500	Withdraw	0
14	10	1009	30000	Deposit	0
15	11	1010	2000	Withdraw	0

14. Calculate running total of transaction amounts using SUM() OVER().

```
select TransactionID, AccountID, TransactionType, Amount,  
sum(Amount) over(partition by AccountID order by TransactionDate) as Running_Total  
from Transactions;
```



	TransactionID	AccountID	TransactionType	Amount	Running_Total
4	12	1002	Withdraw	10000	35000
5	4	1003	Withdraw	3000	3000
6	13	1003	Deposit	5000	8000
7	5	1004	Deposit	15000	15000
8	6	1005	Deposit	12000	12000
9	14	1005	Withdraw	3500	15500
10	7	1006	Withdraw	7000	7000
11	8	1007	Deposit	9000	9000
12	15	1007	Deposit	15000	24000
13	9	1008	Withdraw	4500	4500
14	10	1009	Deposit	30000	30000
15	11	1010	Withdraw	2000	2000

15. Find the second highest account balance using subquery or window function.

```
select *  
from(select AccountID ,Balance, c.CustomerName ,  
    DENSE_RANK() over(order by Balance desc) as Rnk  
    from Accounts a  
    join Customers c on a.CustomerID = c.CustomerID  
    ) as ranked  
where Rnk = 2;
```



Results		Messages		
	AccountID	Balance	CustomerName	Rnk
1	1006	150000	Neha Joshi	2

16. Find customers who performed more than 2 transactions.

```
from(select c.CustomerID, c.CustomerName ,  
    DENSE_RANK() over(partition by c.CustomerID order by TransactionDate) as No_of_Transaction  
from Transactions t  
join Accounts a on t.AccountID = a.AccountID  
join Customers c on a.CustomerID = c.CustomerID  
 ) as RankedData
```



Results		Messages	
	CustomerID	CustomerName	No_of_Transaction
1	1	Rahul Sharma	2
2	2	Sneha Patil	2
3	3	Aman Verma	2
4	5	Karan Mehta	2
5	7	Rohit Kumar	2

17. Display customer-wise minimum and maximum transaction amounts.

```
select c.CustomerName , min(t.Amount) as Min_Amount , max(t.Amount) as Max_Amount
from Customers c
join Accounts a on c.CustomerID = a.CustomerID
join Transactions t on a.AccountID = t.AccountID
group by c.CustomerName
order by Max_Amount desc;
```



	CustomerName	Min_Amount	Max_Amount
1	Vivek Shah	30000	30000
2	Sneha Patil	10000	25000
3	Rohit Kumar	9000	15000
4	Priya Singh	15000	15000
5	Karan Mehta	3500	12000
6	Rahul Sharma	5000	10000
7	Neha Joshi	7000	7000
8	Aman Verma	3000	5000
9	Pooja Sharma	4500	4500
10	Anjali Verma	2000	2000



**THANK
YOU**